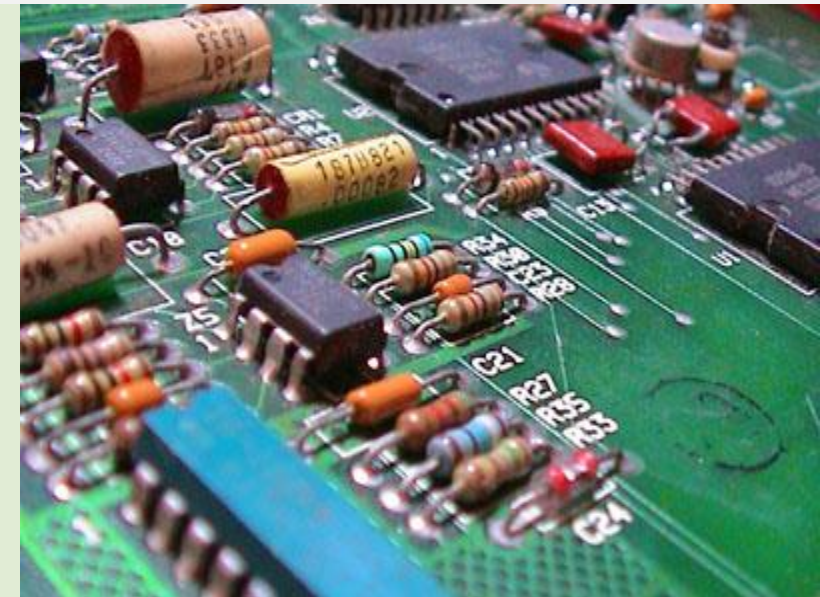




Embedded Systems and Robotics

ICE 372

Lecture 6: Memory Interfacing



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Example

A door sensor is connected to the P1.1 pin, and a buzzer is connected to P1.7. Write an 8051 C program to monitor the door sensor, and when it opens, sound the buzzer. You can sound the buzzer by sending a square wave of a few hundred Hz.

Solution:

```
#include <reg51.h>
void MSDelay(unsigned int);
sbit Dsensor=P1^1;
sbit Buzzer=P1^7;

void main(void)
{
    Dsensor=1;                //make P1.1 an input
    while (1)
    {
        while (Dsensor==1) //while it opens
        {
            Buzzer=0;
            MSDelay(200);
            Buzzer=1;
            MSDelay(200);
        }
    }
}
```

Example

Write an 8051 C program to toggle all the bits of P0, P1, and P2 continuously with a 250 ms delay.

Use the sfr keyword to declare the port addresses.

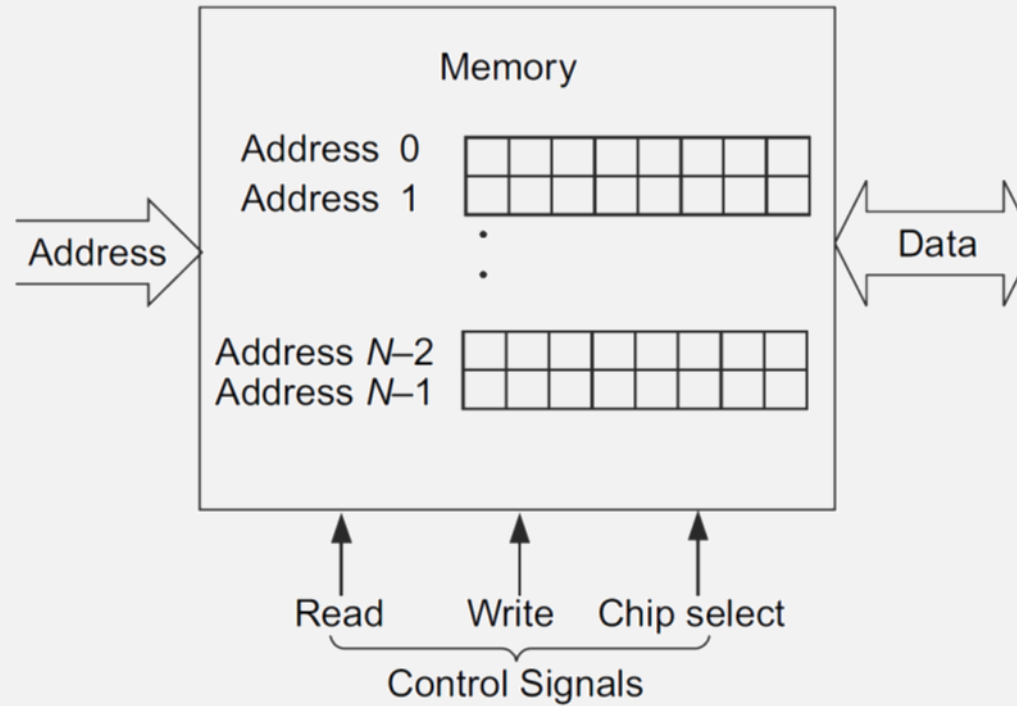
Solution:

```
//Accessing Ports as SFRs using sfr data type
sfr P0=0x80;
sfr P1=0x90;
sfr P2=0xA0;
void MSDelay(unsigned int);

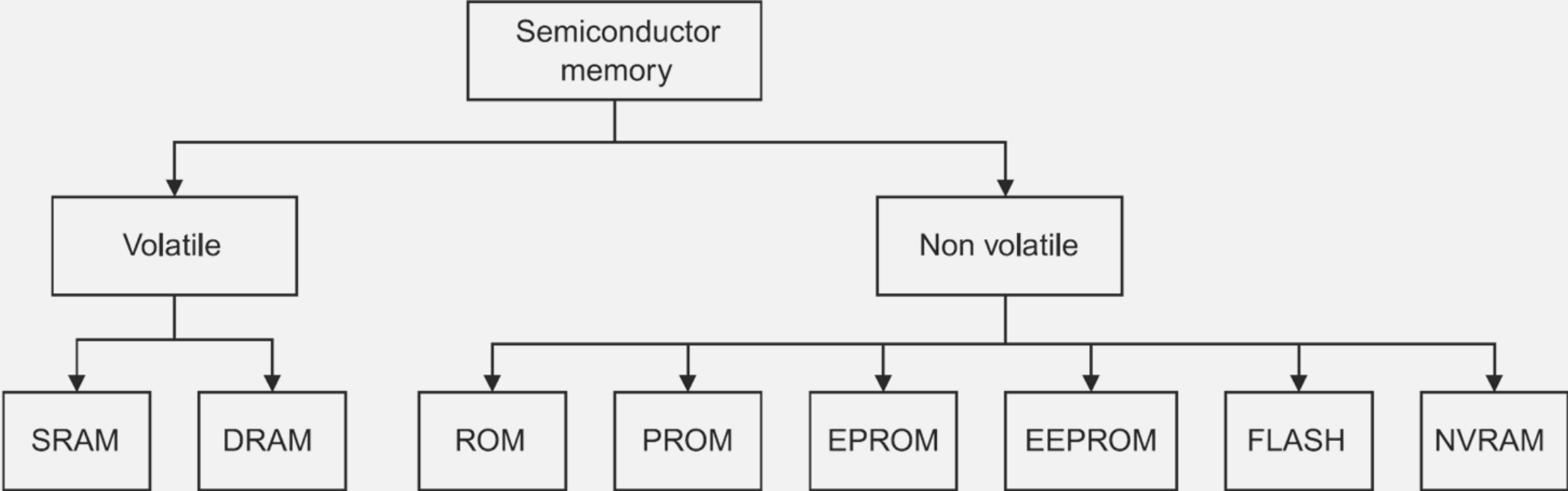
void main(void)
{
    while (1)
    {
        P0=0x55;
        P1=0x55;
        P2=0x55;
        MSDelay(250);
        P0=0xAA;
        P1=0xAA;
        P2=0xAA;
        MSDelay(250);
    }
}
```

Memory Chips

- ❑ All the memory devices have a common set of input and output signals like address, data and control signals.



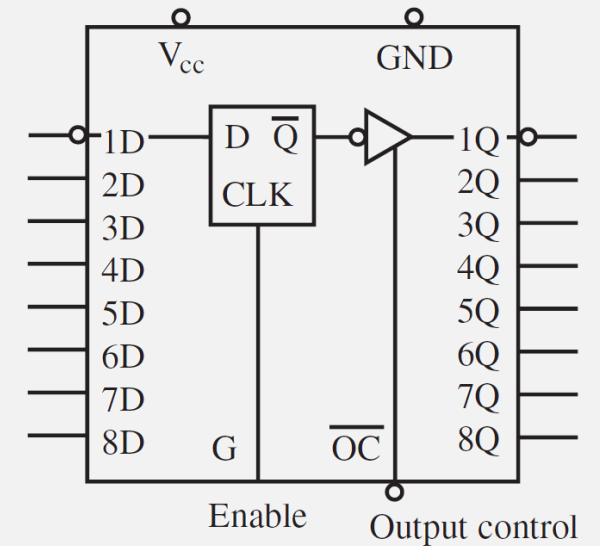
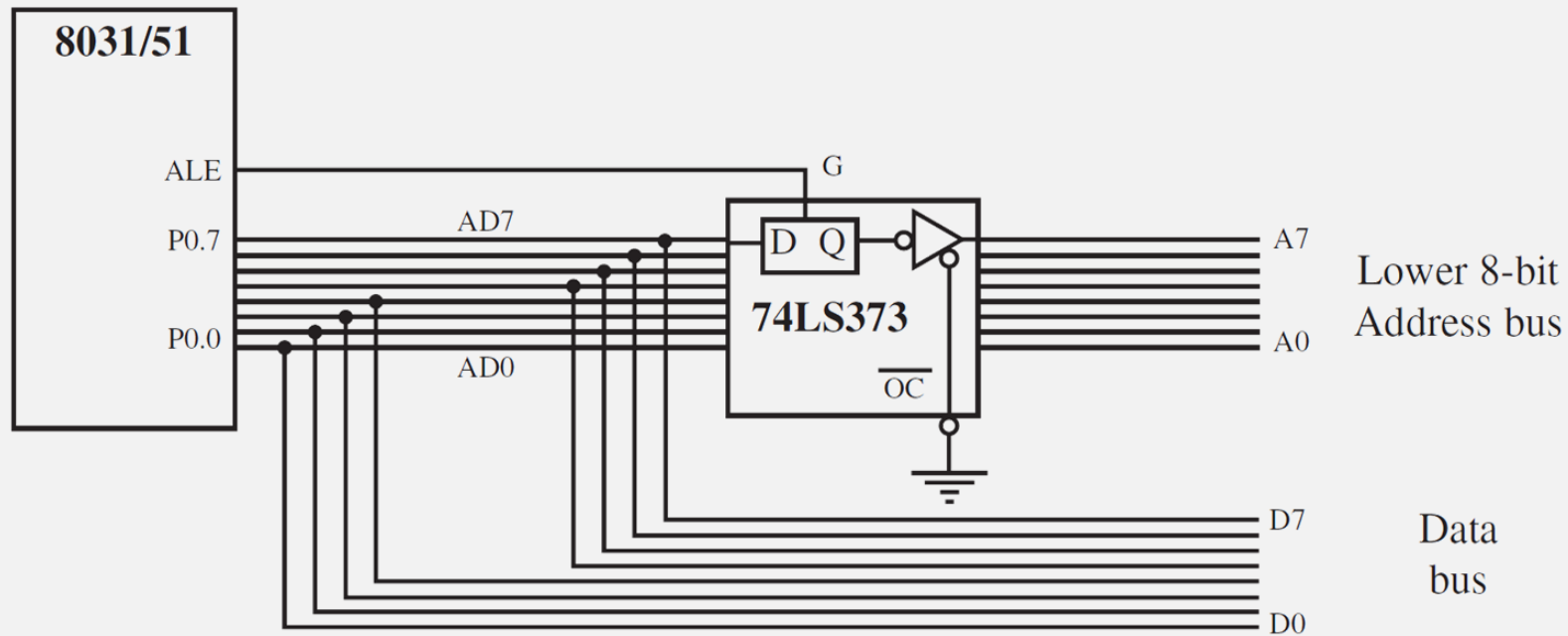
Semiconductor Memory Devices



❑ Port 0 is also designated as AD0-AD7, allowing it to be used for both address and data, when connecting an 8051/31 to an external memory, port 0 provides both address and data. The 8051 multiplexes address and data through port 0 to save pins.

ALE indicates if P0 has address or data

- When ALE=0, it provides data D0-D7
- When ALE=1, it has address A0-A7

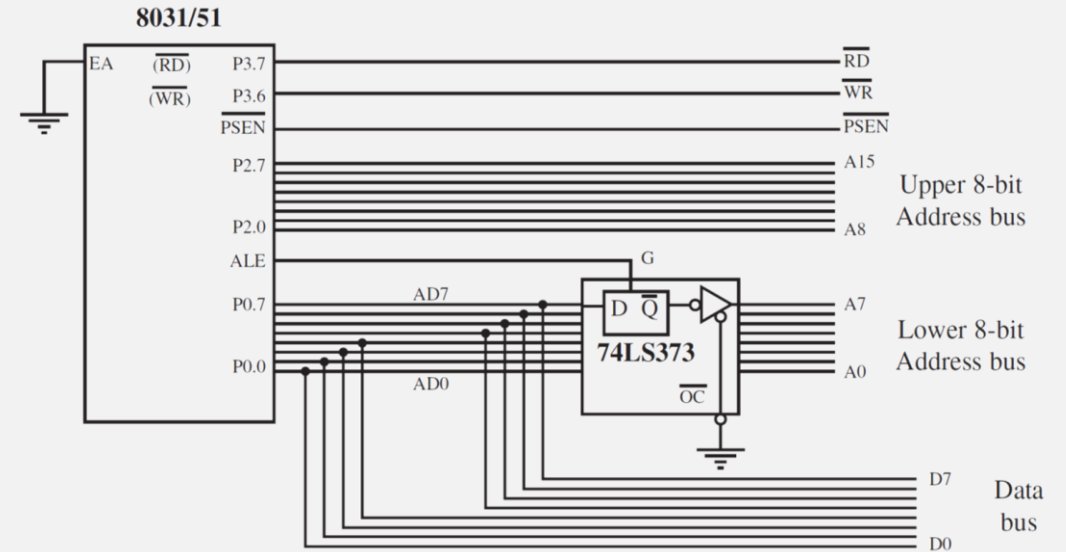
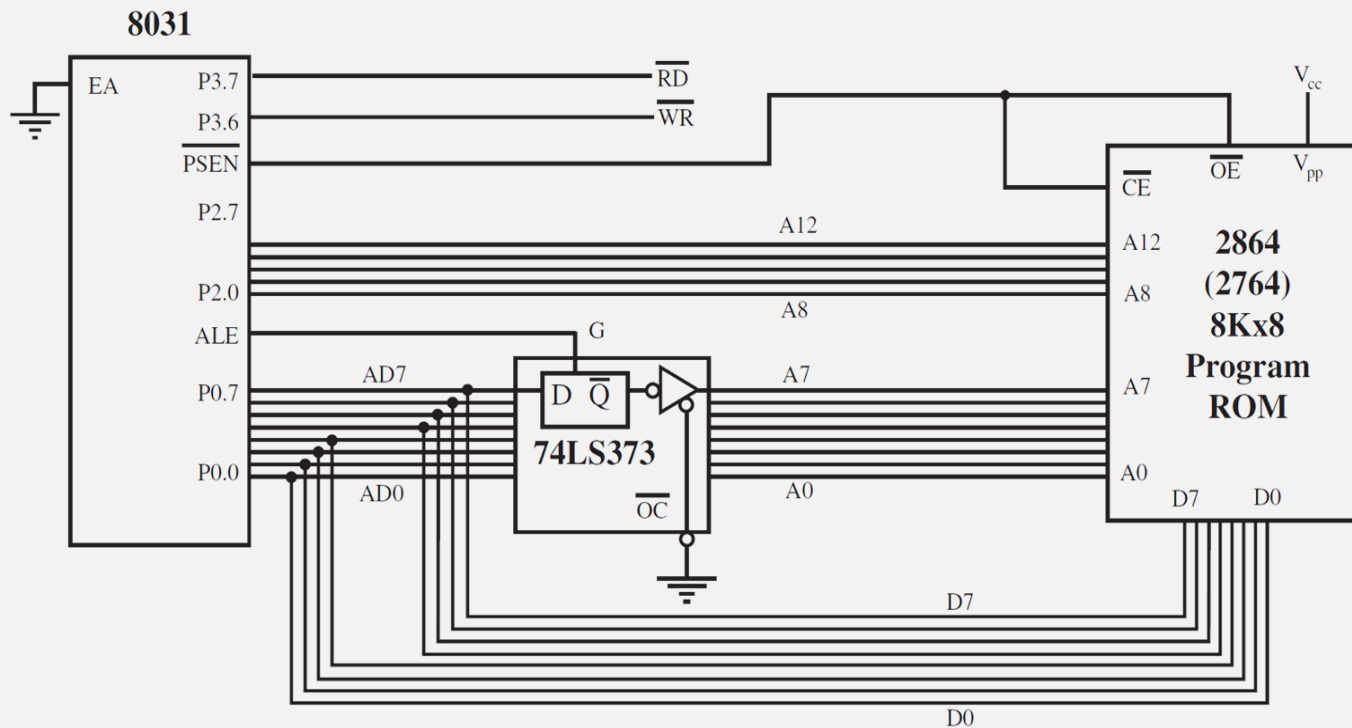


Funtion Table

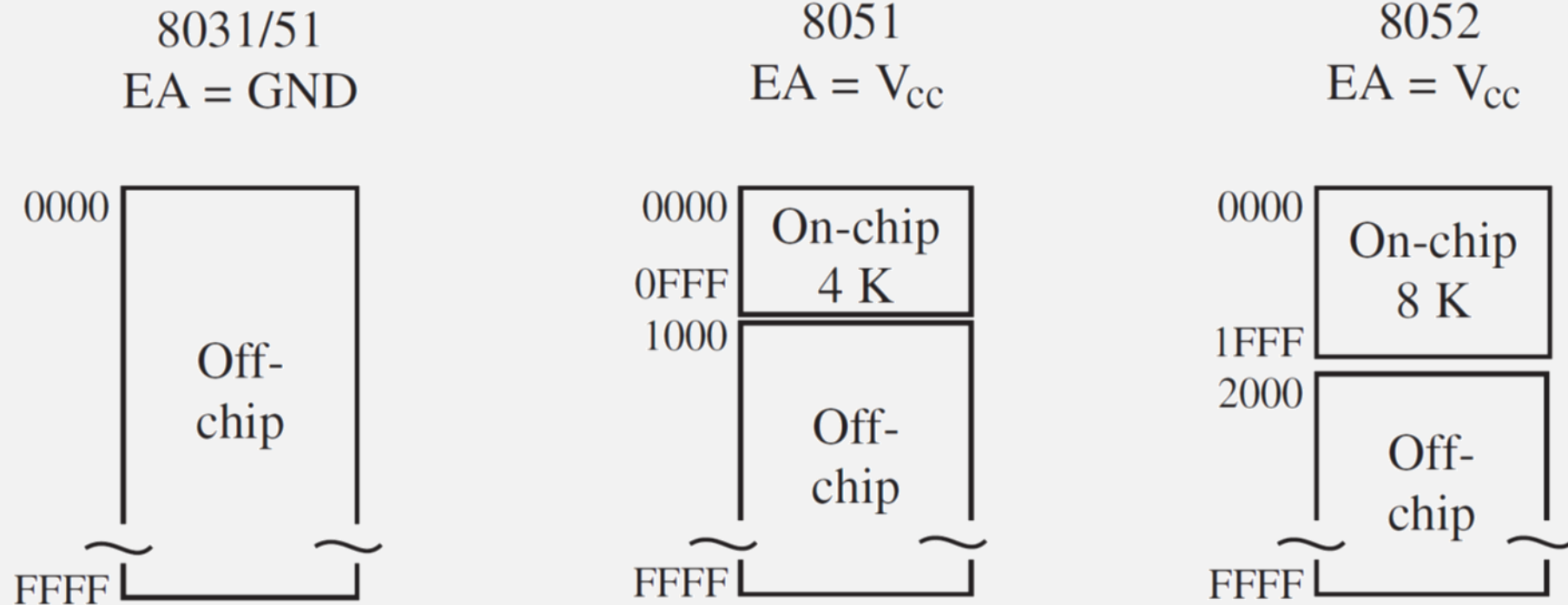
Output control	Enable		Output
	G	D	
L	H	H	H
L	H	L	L
L	L	X	Q0
H	X	X	Z

PSEN (Program Store Enable)

- ❑ PSEN is an output signal for the 8031/51 microcontroller and must be connected to the OE pin of a ROM containing the program code. In other words, to access external ROM containing program code, the 8031/51 uses the PSEN signal.

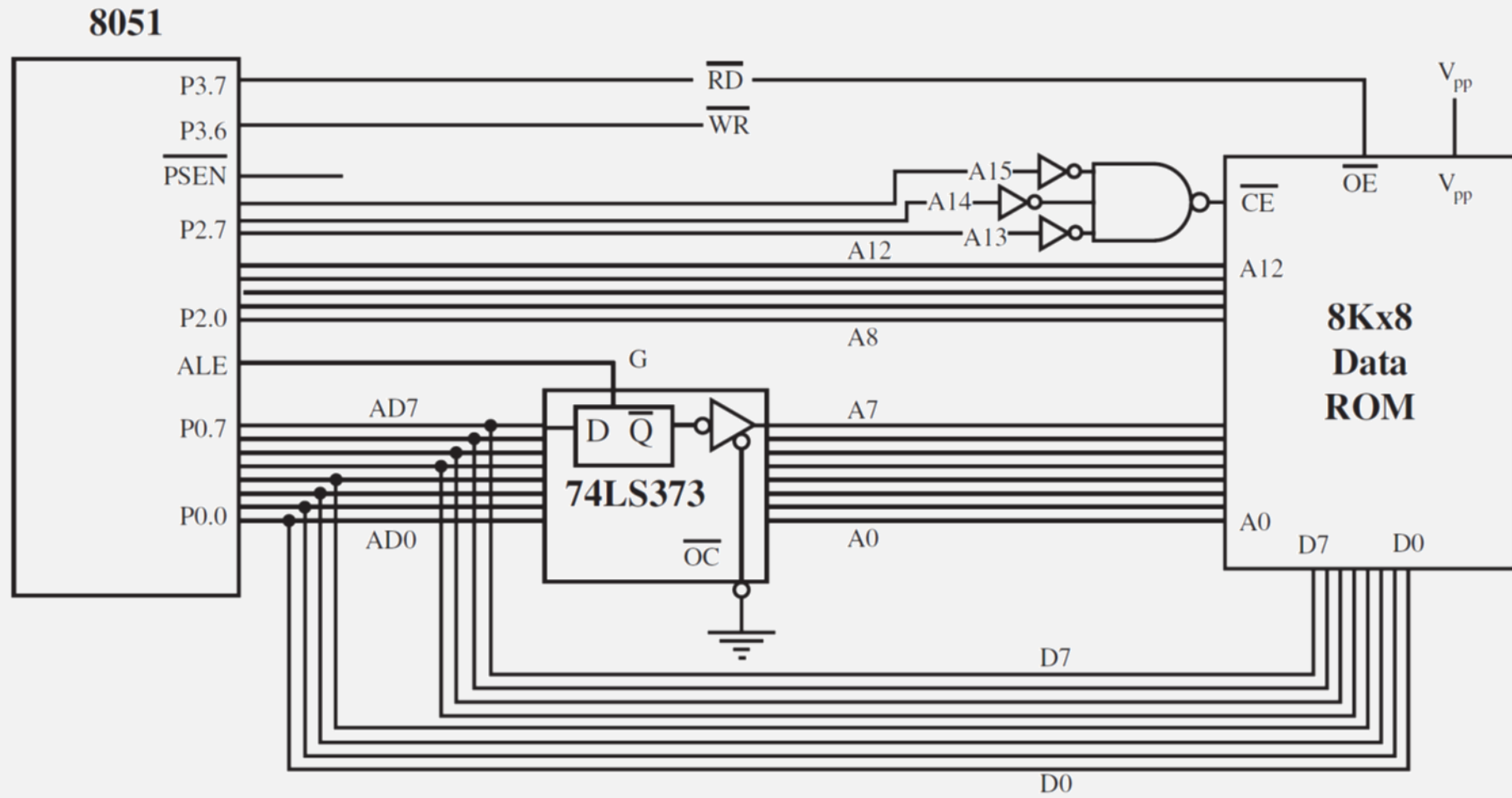


- ❑ For the system that could use the on-chip ROM for the boot code, and an external ROM (using NV-RAM) we still have $EA = V_{CC}$, meaning that upon reset the 8051 executes the on-chip program first; then, when it reaches the end of the on-chip ROM it switches to external ROM for the rest of the program code.



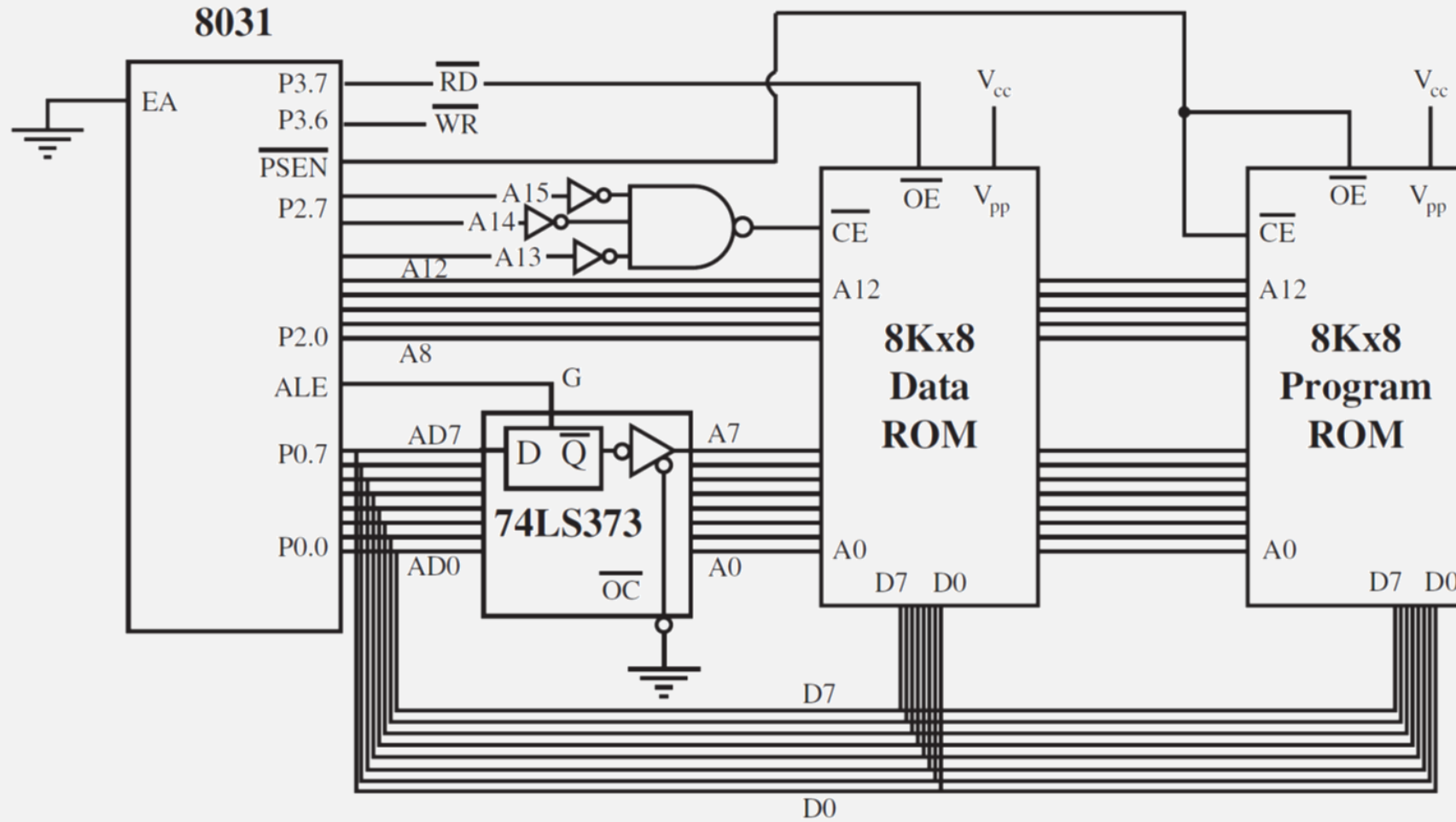
8051 Data Memory Space

- ❑ The 8051 has 128K bytes of address space of which 64K bytes are set aside for program code and the other 64K bytes are set aside for data.
- ❑ Program space is accessed using the program counter to locate and fetch instructions, but the data memory space is accessed using the DPTR register and an instruction called MOVX, where X stands for external.
- ❑ To connect the 8031/51 to external ROM containing data, we use RD (pin P3.7).
- ❑ For the ROM containing the program code, PSEN is used to fetch the code. For the ROM containing data, the RD signal is used to fetch the data. To access the external data memory space, we must use the instruction MOVX.

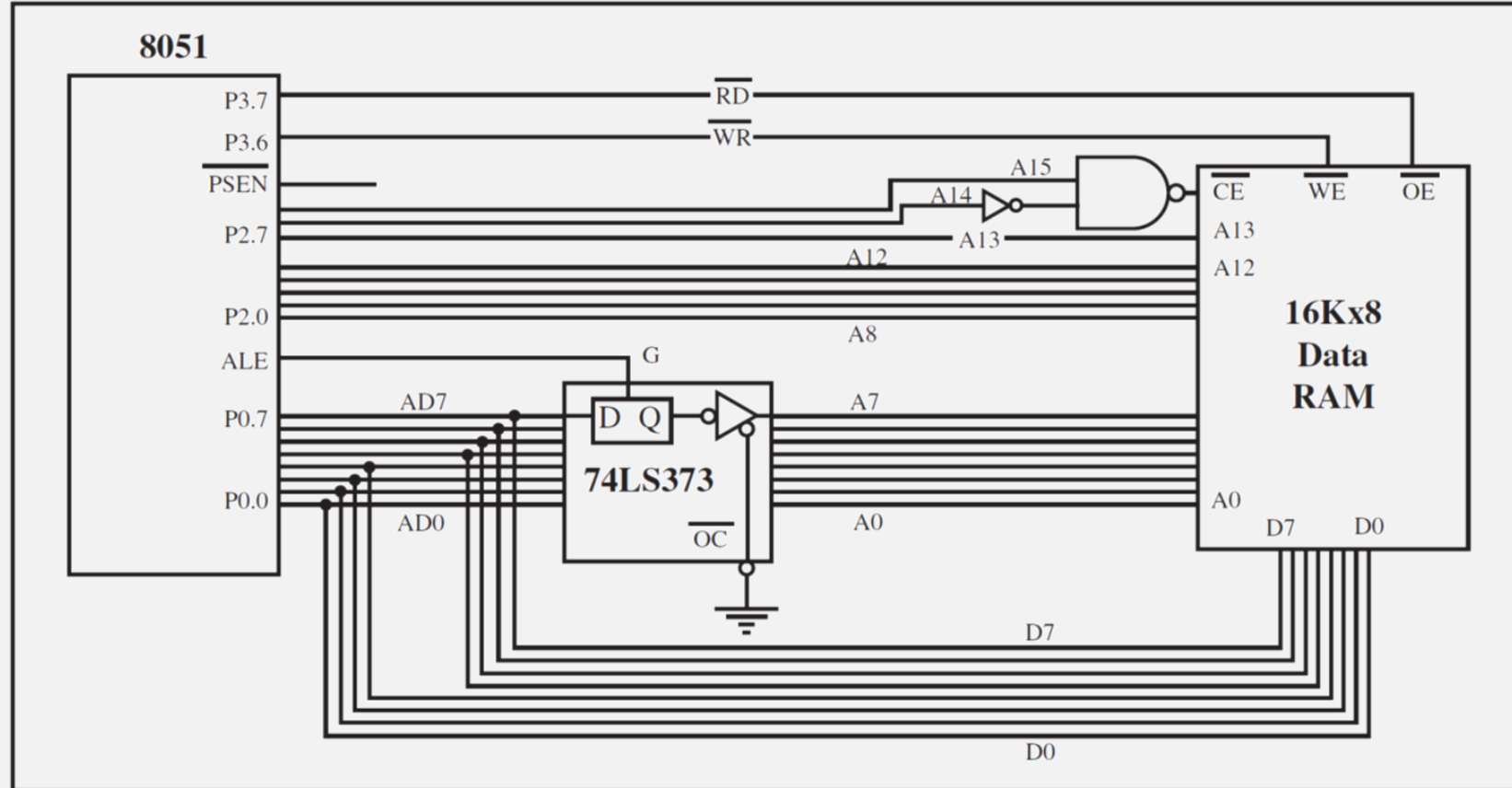


Example

Show the design of an 8031-based system with 8K bytes of program ROM and 8K bytes of data ROM.

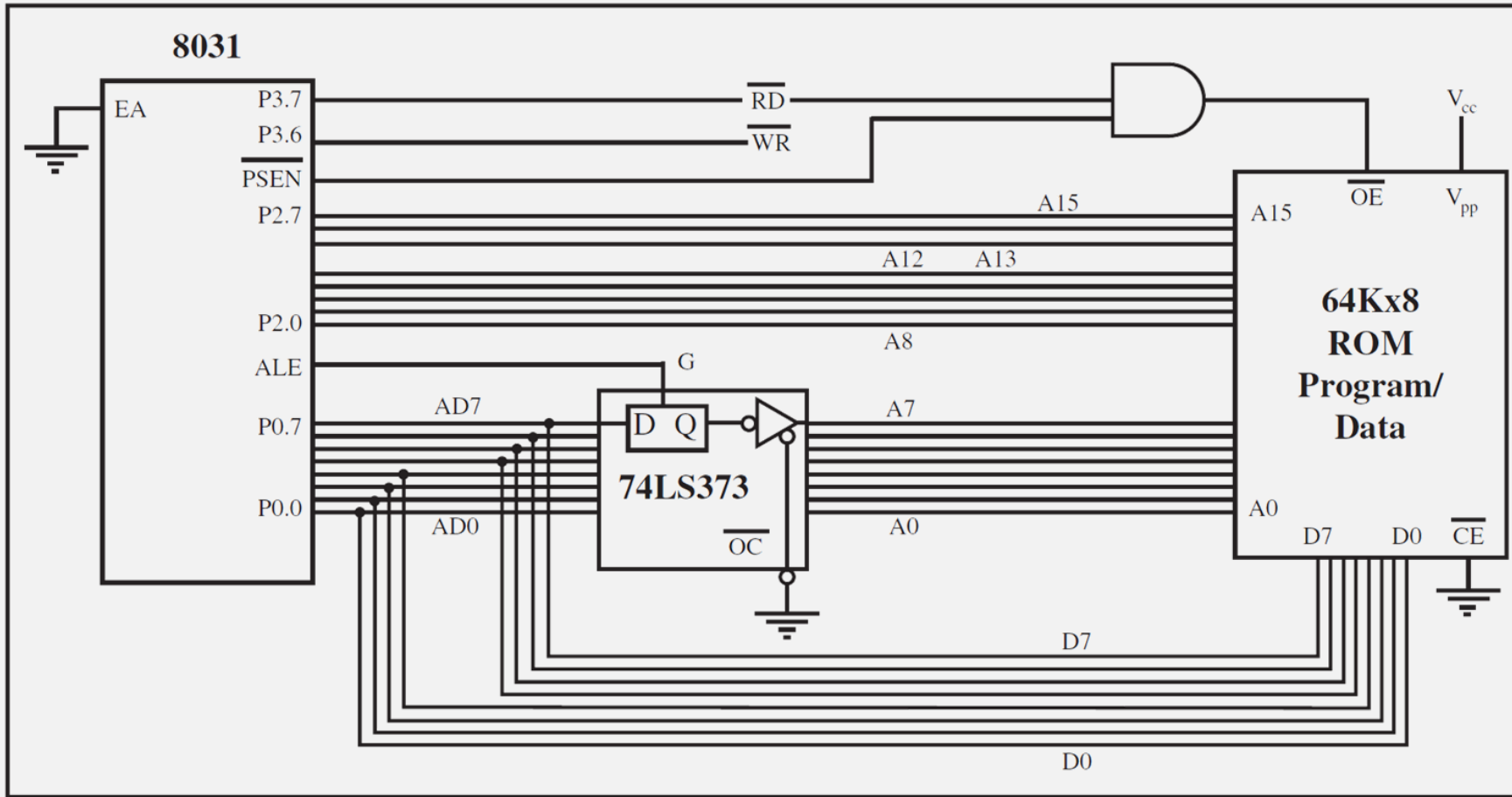


External data RAM



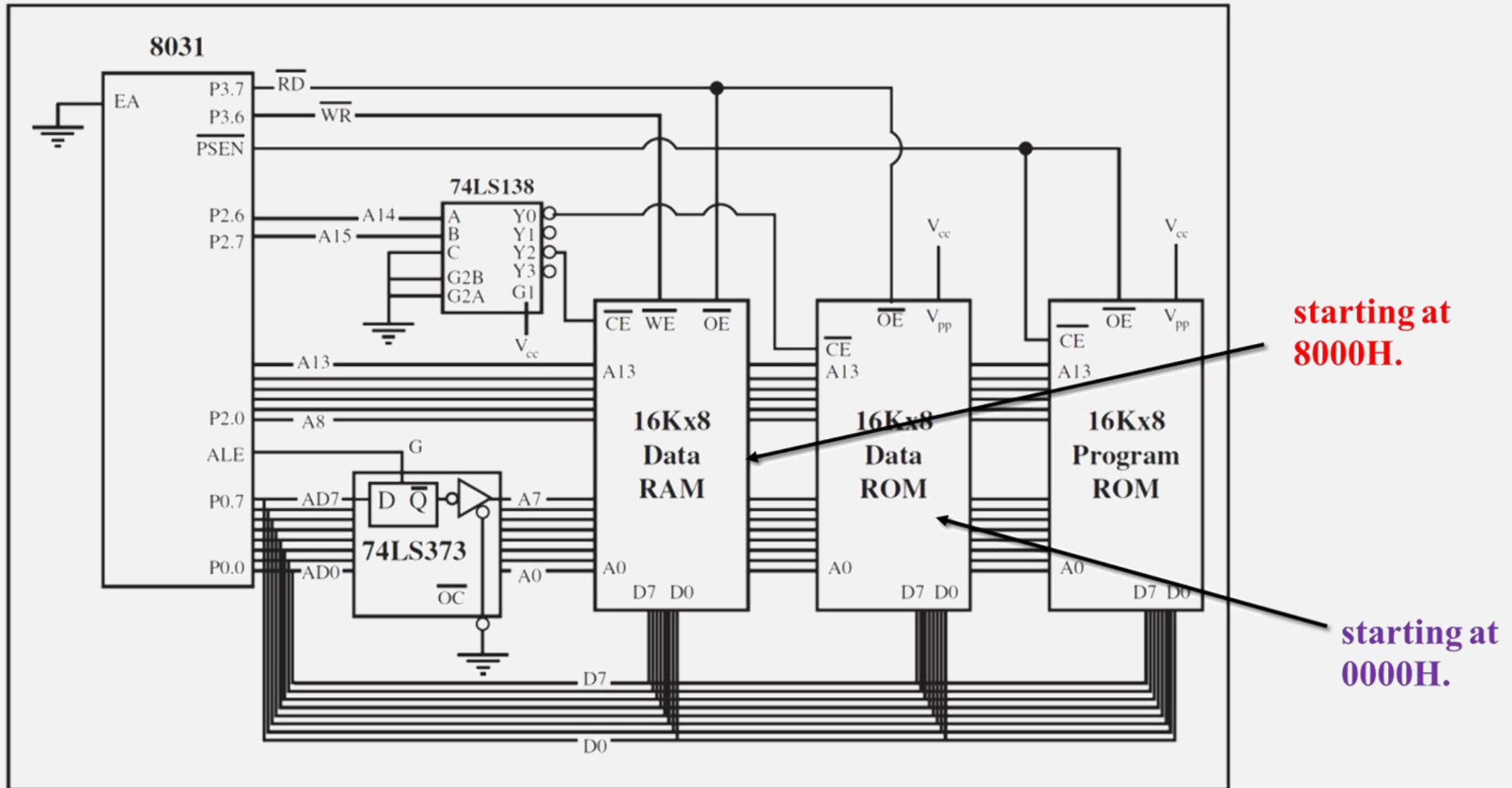
❑ To connect the 8051 to an external SRAM, we must use both RD (P3.7) and WR (P3.6).

A single external ROM for code and data



- ❑ To allow a single ROM chip to provide both program code space and data space, we use an AND gate to signal the OE pin of the ROM chip.

Connection to External Program ROM, Data RAM, and Data ROM



- ❑ To access the external data space (RAM or ROM) of the 8051 using C, we use XBYTE[loc] where loc is an address in the range of 0000–FFFFH.

Example 12

Write a C program (a) to store ASCII letters 'A' to 'E' in external RAM addresses starting at 0, then (b) get the same data from the external RAM and send it to P1 one byte at a time.

Solution:

```
#include <reg51.h>
#include <absacc.h> //notice the header file for XBYTE

void main(void)
{
    unsigned char x;
    XBYTE[0]='A'; //write ASCII 'A' to External RAM location 0
    XBYTE[1]='B'; //write ASCII 'B' to External RAM location 1
    XBYTE[2]='C'; //write ASCII 'C' to External RAM location 2
    XBYTE[3]='D';
    XBYTE[4]='E';
    for(x=0;x<5;x++)
        P1=XBYTE[x]; //read external RAM data and send it to
P1
}
```

Example 13

An external ROM uses the 8051 data space to store the look-up table (starting at 100H) for DAC data. Write a C program to read 30 bytes of table data and send it to P1.

Solution:

```
#include <reg51.h>
#include <absacc.h> //notice the header file for XBYTE

void main(void)
{
    unsigned char count;
    for(count=0;count<30;count++)
        P1=XBYTE[0x100+count];
}
```